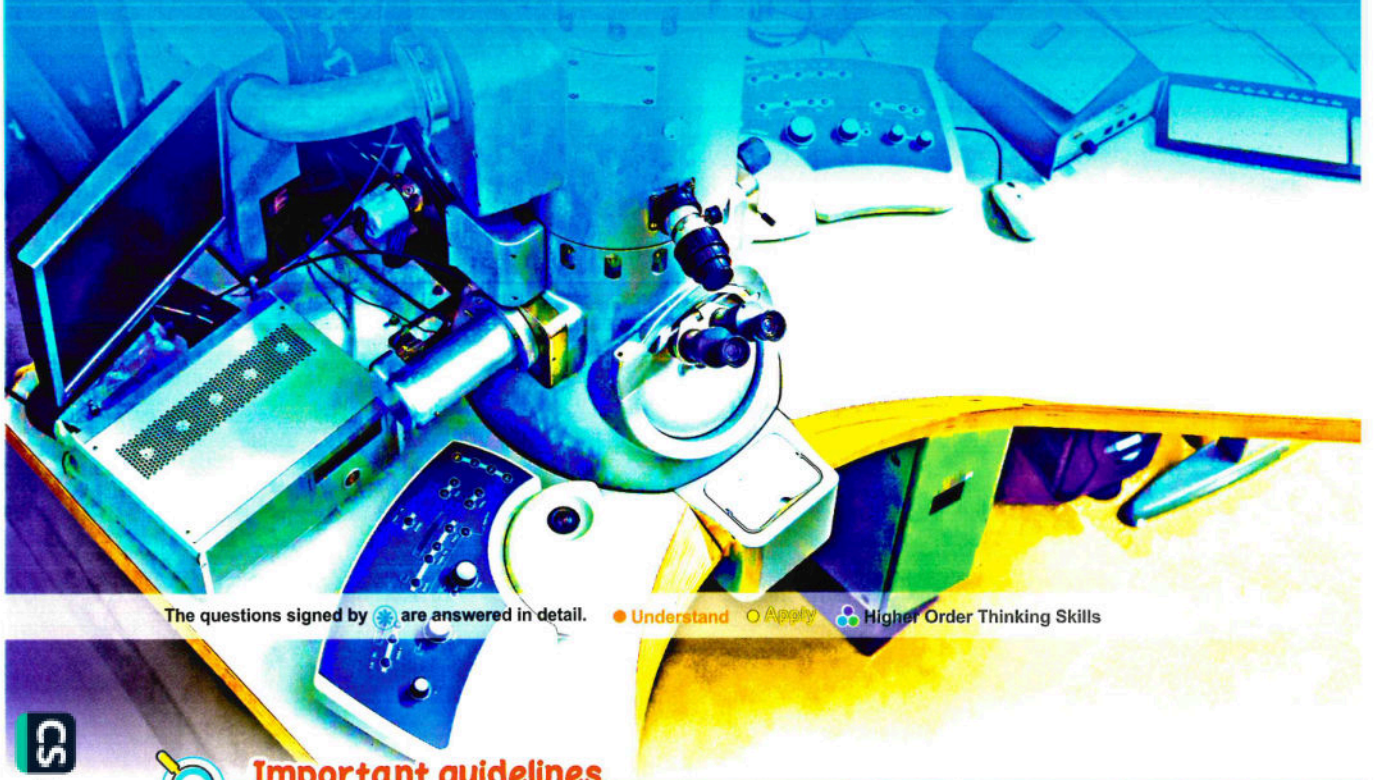
Chapter 5

Questions on Lesson Two

- Compton Effect.
- Wave Particle Duality.
- Electron Microscope.

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

Photon

Energy	$E = h\upsilon = \frac{hc}{\lambda} = mc^2 = P_L c$
Frequency	$\upsilon = \frac{c}{\lambda} = \frac{E}{h}$
Equivalent mass	$m = \frac{E}{c^2} = \frac{h\upsilon}{c^2} = \frac{h}{\lambda c}$
Linear momentum	$P_L = \frac{E}{c} = \frac{h\upsilon}{c} = \frac{h}{\lambda} = mc$
The power of light ray	$P_w = E\phi_L = h\upsilon\phi_L = \frac{hc}{\lambda}\phi_L = P_L c\,\phi_L$
The acting force of a light ray on a surface	$F = \frac{2P_w}{c} = \frac{2h\upsilon\phi_L}{c} = \frac{2h\phi_L}{\lambda}$

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De-Broglie equation

Particle

$$\lambda = \frac{h}{P_L} = \frac{h}{mv}$$

The electron

The wave associated with its motion	The current that it produces	When accelerating it
$\lambda = \frac{h}{P_L} = \frac{h}{m_e v}$	$I = \frac{Ne}{t} = \phi_L e$	$eV = KE = \frac{1}{2} m_e v^2$

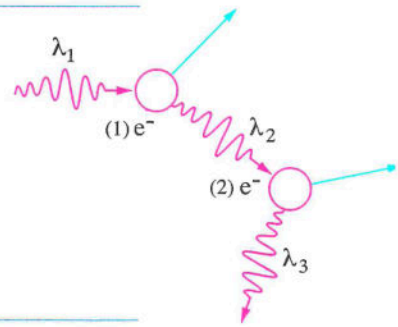
First Multiple choice questions

Use the following constants if needed:

$(c = 3 \times 10^8$ m/s, $h = 6.625 \times 10^{-34}$ J.s, $m_e = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C)

1. In Compton effect, the ratio of the electron's mass before collision to its mass after collision is ..... .
   (a) greater than one
   (b) less than one
   (c) equal to one
   (d) determined by the mass of the photon

2. The opposite figure shows X-ray photon that collides with two free electrons, where the photon has collided with the first electron then the scattered photon has collided with the second electron, so ..... .



   (a) $\lambda_1 > \lambda_2 > \lambda_3$
   (b) $\lambda_1 < \lambda_2 > \lambda_3$
   (c) $\lambda_1 > \lambda_2 < \lambda_3$
   (d) $\lambda_1 < \lambda_2 < \lambda_3$

3. Compton effect proves ..... .
   (a) the wave nature of photons
   (b) the particle nature of electron
   (c) the particle nature of photons
   (d) the wave nature of electron

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4 In Compton effect, when a high frequency photon collides with a free electron, which of the following quantities increases for the photon after collision?

(a) Energy. (b) Frequency. (c) Wavelength. (d) Momentum.

5 In Compton effect after the collision, the quantity that **doesn't** decrease is ..... .

(a) the wavelength associated with the electron

(b) the photon's energy

(c) the photon's frequency

(d) the photon's speed

6 In Compton effect, when X-ray photon collides with a free electron, which of the following has less speed?

(a) The photon before collision (b) The electron before collision

(c) The photon after collision (d) The photon after collision

7 The ratio between the energy of photon after collision and its energy before collision in Compton effect is ..... .

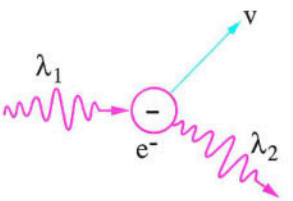
(a) greater than one (b) equal to one (c) less than one (d) indeterminable

8 In Compton effect, if the energy of the incident photon (before collision) is (E_1), the energy of the scattered photon (after collision) is (E_2) and the kinetic energy gained by electron as a result of collision is (E_3), which of the following relations is correct?

(a) $E_3 = E_1 + E_2$ (b) $E_3 = E_1 - E_2$ (c) $E_3 < E_1 - E_2$ (d) $E_3 > E_1 - E_2$

9 The opposite diagram represents Compton scattering in which the photon's energy is changed by ΔE.

If Planck's constant is h and the speed of light is c, the quantity $\frac{hc}{\Delta E}$ equals ..... .



(a) $\frac{\lambda_2 - \lambda_1}{\lambda_2 \lambda_1}$ (b) $\frac{\lambda_2 + \lambda_1}{\lambda_2 \lambda_1}$ (c) $\frac{\lambda_2 \lambda_1}{\lambda_2 - \lambda_1}$ (d) $\frac{\lambda_2 \lambda_1}{\lambda_2 + \lambda_1}$

10 In Compton effect, as the difference between the wavelength of the incident photon and the wavelength of the scattered photon increases with the constancy of all factors, the energy gained by the electron ..... .

(a) increases (b) decreases

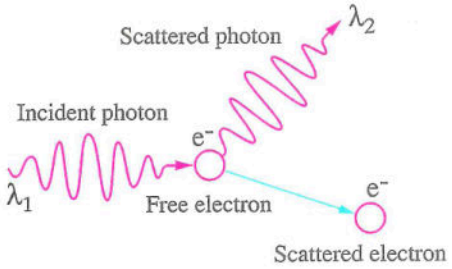
(c) doesn't change (d) cannot be determined

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11 X-ray photon of wavelength 7 Å has collided with a free electron. The kinetic energy of the electron increased by 1.853×10^{-18} J, then the wavelength of the scattered X-ray photon equals ..... .

(a) 5.6 Å (b) 7.46 Å (c) 7.046 Å (d) 8.2 Å

12 The opposite figure illustrates Compton's phenomenon. If the ratio between the wavelength of the incident photon and that of the scattered photon $\left(\frac{\lambda_1}{\lambda_2}\right)$ equals $\frac{999}{1000}$, the photon after being scattered loses ..... from its energy before collision.



(a) $\frac{1}{999}$ (b) $\frac{1}{1000}$

(c) $\frac{999}{1000}$ (d) $\frac{1000}{999}$

13 In Compton effect, if the photon loses X% from its energy as a result of collision, ...

(a) the wavelength of the photon increases by X%

(b) the momentum of photon decreases by X %

(c) the mass of electron increases by X %

(d) the kinetic energy of electron decreases by X %

14 * The equivalent mass for a photon of electromagnetic waves whose wavelength is 100 nm equals ..... .

(a) 4.42×10^{-31} kg (b) 4.42×10^{-32} kg (c) 2.2×10^{-34} kg (d) 2.21×10^{-35} kg

15 If a mass of 1.6749×10^{-27} kg is converted into energy, this produced energy equals ..... .

(a) 1.7×10^{-10} J (b) 1.5×10^{-10} J

(c) 9.1×10^{-31} J (d) 3.4×10^{8} J

16 From the properties of the photon, ...

(a) its speed in space equals 3×10^{8} m/s (b) it can be accelerated

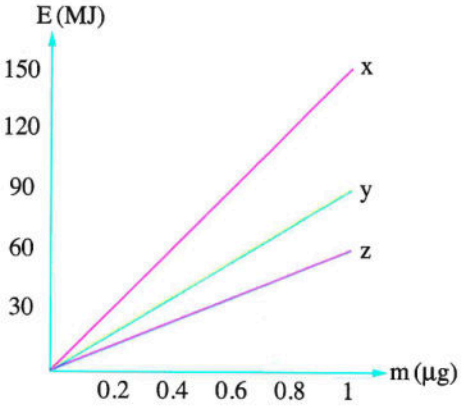
(c) it deflects by the electric field (d) all the previous

17 The ratio between the frequencies of two photons is 1 : 2, so the ratio between their energies is ..... respectively.

(a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 4 : 1

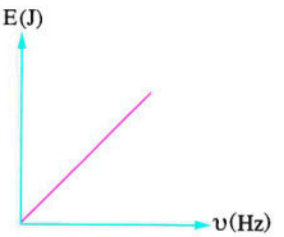
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18 Assuming a mass of 1 μg gradually converts into energy, which of the graphical lines x, y or z in the opposite figure represents the relationship between the resulting energy (E) and the transformed mass (m)?



(a) Line x
(b) Line y
(c) Line z
(d) The three lines.

19 The opposite graph represents the relation between the energies of photons (E) and their frequencies (υ). So, the slope of the straight line equals ..... .



(a) the wavelength (λ) (b) Planck's constant (h)
(c) light's speed (c) (d) linear momentum (P_L)

20 * Consider a small town consumes energy of 10^8 J every second, then the required amount of mass to be converted completely into energy in a nuclear reactor to supply this town with energy for one year (365.25 days) equals ..... .

(a) 0.035 g (b) 0.58 g (c) 1.46 g (d) 35 g

21 A photon of momentum P_L, then its energy equals ..... . (Where: c is the speed of light)

(a) $P_L c$ (b) $\frac{P_L c}{2}$ (c) $\frac{P_L^2}{2c}$ (d) $\frac{P_L c}{4}$

22 A photon of wavelength (λ) and frequency (υ). So, its linear momentum is ..... .

(a) $\frac{h}{\lambda}$ (b) $\frac{h\upsilon}{\lambda}$ (c) $\frac{hc}{\lambda}$ (d) $\frac{h\upsilon}{c^2}$

23 The number of photons in a monochromatic light beam of wavelength λ is n, so the total energy of the beam equals ..... .

(a) $\frac{hc}{n\lambda}$ (b) $\frac{n\lambda}{hc}$ (c) $\frac{nhc}{\lambda}$ (d) $\frac{nc}{\lambda}$

24 The ratio between the linear momentum of a photon and its equivalent mass equals ..... .

(a) photon's speed (b) Planck's constant (c) photon's energy (d) its wavelength

25 When the intensity of a monochromatic light ray doubles, the momentum for each photon ..... .

(a) decreases to the half (b) is doubled
(c) increases four times (d) does not change

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26 ✳ A photon of wavelength 770 nm, then:

(i) Its energy equals ..... .

(a) 2.58×10^{-22} J (b) 2.58×10^{-21} J (c) 2.58×10^{-20} J (d) 2.58×10^{-19} J

(ii) The equivalent of its mass equals ..... .

(a) 2.87×10^{-37} kg (b) 2.87×10^{-36} kg (c) 2.58×10^{-38} kg (d) 2.58×10^{-35} kg

(iii) Its momentum equals ..... .

(a) 8.6×10^{-28} kg.m/s (b) 8.6×10^{-29} kg.m/s

(c) 7.7×10^{-30} kg.m/s (d) 7.7×10^{-29} kg.m/s

27 Two photons (x) and (y) propagate in air, photon (x) of linear momentum (P_L) and photon (y) of linear momentum ($2\ P_L$), then ..... .

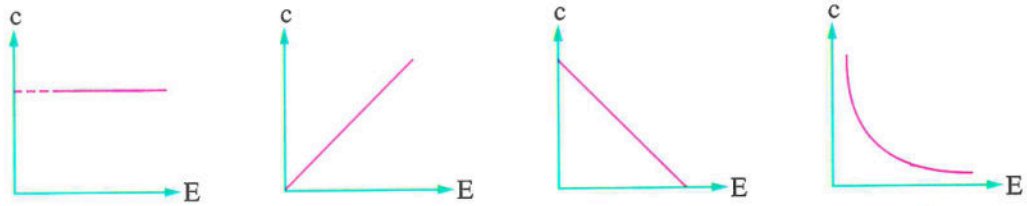
(a) the wavelength of photon (x) is double the wavelength of photon (y)

(b) the wavelength of photon (x) is half the wavelength of photon (y)

(c) the speed of photon (x) is double the speed of photon (y)

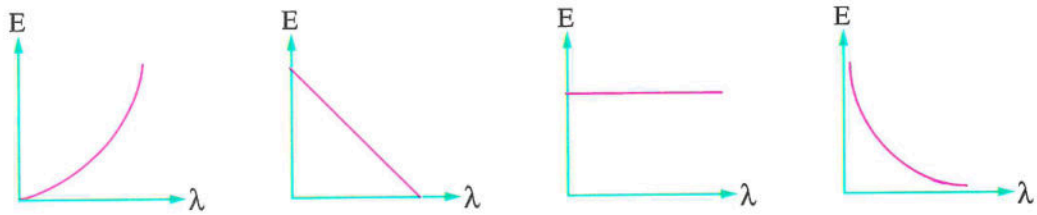
(d) the speed of photon (x) is half the speed of photon (y)

28 Which of the following graphs represents the relation between the speed (c) of photons propagate in air and the energy (E) for each of them?



(a) (b) (c) (d)

29 Which of the following graphs represents the relation between the energy of the photon (E) and its wavelength (λ)?



(a) (b) (c) (d)

30 A monochromatic light beam falls on a certain area for a certain time. If the intensity of that beam is doubled, so when it falls on the same area for the same time interval, ...

(a) the energy of the photon is doubled

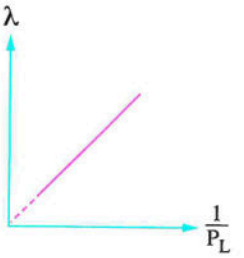
(b) the linear momentum of the photon is doubled

(c) the equivalent mass of the photon decreases to its half

(d) the number of photons is doubled

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31 The opposite graph represents the relation between the wavelength (λ) for different beams of light and the reciprocal of the linear momentum $\left(\frac{1}{P_L}\right)$ of photons in each beam. So, the slope of the straight line equals ..... .



(a) light speed (b) Planck's constant
(c) the mass of the photon (d) the photon's frequency

32 ✳ A potential difference is applied to an electron, therefore its momentum has increased by 25%, so its kinetic energy increases approximately by ..... .

(a) 65 % (b) 56 % (c) 38 % (d) 25 %

33 ✳ A light ray whose photon's energy is E has fallen perpendicularly on a reflecting surface, so the amount of change in the momentum of the photon when it reflects equals ..... . (Where: (c) is the speed of light)

(a) $\frac{E}{c}$ (b) $\frac{2E}{c}$ (c) $\frac{2E}{c^2}$ (d) $\frac{E}{c^2}$

34 Photons from a radiation of wavelength λ fall on a surface, if the intermolecular distance between the surface particles is in range 0.01 λ, most of these photons ..... .

(a) interfere at the surface (b) pass through the surface straight away.
(c) are refracted inside the surface (d) are reflected from the surface

35 A beam of monochromatic light of wavelength λ falls perpendicularly on a reflecting surface, so the number of the rebounded photons from the surface in one second is (ϕ_L), then the acting force on the surface equals ..... .

(a) $\frac{2hc}{\lambda}\phi_L$ (b) $\frac{2h\lambda}{c}\phi_L$ (c) $\frac{2\lambda c}{h}\phi_L$ (d) $\frac{2h}{\lambda}\phi_L$

36 If a monochromatic light ray has rebounded perpendicularly off a reflecting surface at a rate of 10^{20} photons/s and acted on the surface with a force of 2×10^{-7} N, then the frequency of that light equals ..... .

(a) 7.2×10^{-16} Hz (b) 2.7×10^{16} Hz (c) 3.75×10^{14} Hz (d) 4.5×10^{14} Hz

37 A yellow light beam of total power 1 W and frequency 5.2×10^{14} Hz is reflected from a surface, then the total number of photons reflected from the surface in one second equals ..... .

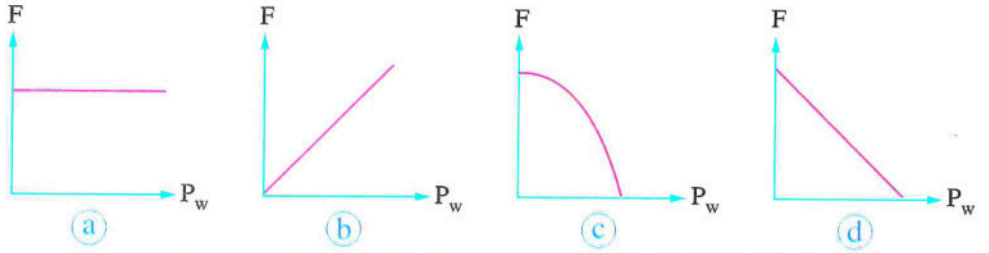
(a) 5.2×10^{20} photons (b) 3.4×10^{19} photons
(c) 2.9×10^{18} photons (d) 6.4×10^{17} photons

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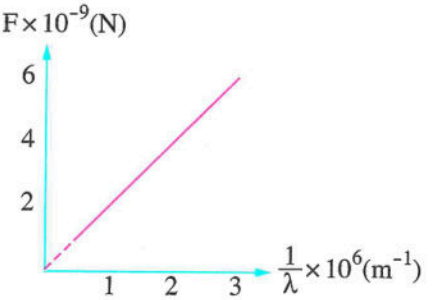
38 Two sources (a) and (b) whose powers are P_w and $2P_w$, emit monochromatic electromagnetic radiations of wavelengths λ, 4λ respectively, so the ratio between the number of photons emitted from the two sources during the same time $\left(\frac{N_a}{N_b}\right)$ equals ..... .

(a) $\frac{8}{1}$ (b) $\frac{4}{1}$ (c) $\frac{1}{8}$ (d) $\frac{1}{4}$

39 Many light beams, each of single wavelength, fall perpendicularly on a reflective surface, which of the following graphs represents the relation between the force (F) by which each beam acts on the surface and the power (P_w) for each of them?



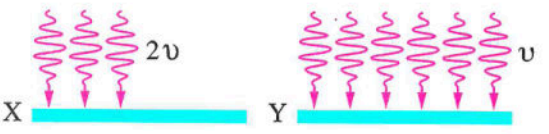
40 Many beams of electromagnetic radiation of single wavelength fall perpendicularly at the same rate (ϕ_L) on a reflective surface and the opposite graph represents the relation between the force (F) by which each beam acts on the surface and the reciprocal of the wavelength ($\frac{1}{\lambda}$) of the photons of that beam, so the falling rate of photons of these light beams (ϕ_L) equals ..... .

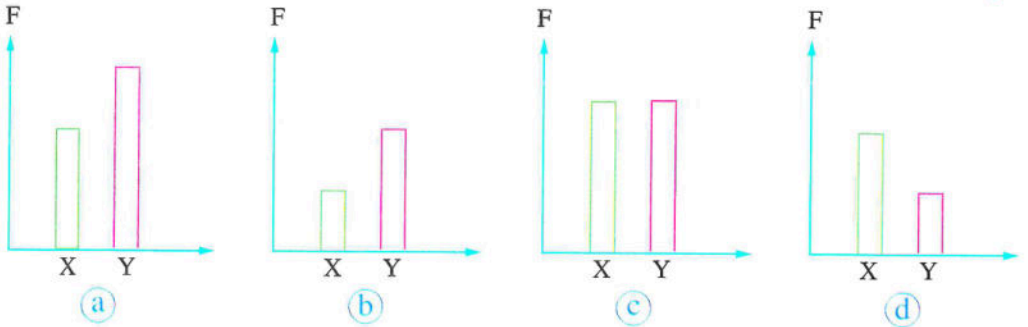


(a) 7.5×10^{17} photons/s (b) 1.5×10^{18} photons/s

(c) 3×10^{18} photons/s (d) 6×10^{18} photons/s

41 The opposite figure represents two light beams of the same power but different in frequency have fallen on two reflecting surfaces X and Y. Which of the following graphs represents the relative forces by which the two beams act on the surface when they are reflected from it?





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2 ✳ An electromagnetic beam of power 10 W reflects perpendicularly from the surface of a reflecting object of mass 10 kg, then the force by which the beam acts on the surface is ..... .

(a) 6.67×10^{-7} N (b) 3.3×10^{-4} N (c) 6.67×10^{-8} N (d) 6.67×10^{-3} N

3 ✳ A radio station emits a wave of frequency 92.4 MHz, then:

(i) The energy of each photon emitted from this station equals ..... .

(a) 6.12×10^{-27} J (b) 6.12×10^{-26} J (c) 8.16×10^{-27} J (d) 8.16×10^{-26} J

(ii) The rate of the emitted photons per second, if the power of the station is 100 kW is ..... .

(a) 1.63×10^{33} photons/s (b) 1.63×10^{32} photons/s

(c) 1.63×10^{30} photons/s (d) 1.63×10^{28} photons/s

4 ✳ A light ray of wavelength 6000 Å has fallen on a metallic surface with a power of 39.6 W, if 1% only of the incident photons liberate electrons, the number of electrons which are liberated from the metallic surface in one second equals ..... electrons.

(a) 12×10^{16} (b) 1.2×10^{18} (c) 12×10^{18} (d) 1.2×10^{15}

5 All the following are from the properties of the electron, **except** ..... .

(a) has a wave nature during its motion

(b) has a particle nature

(c) the wavelength associated with the electron increases by increasing its speed

(d) its kinetic energy increases by increasing its speed

6 If the mass of a moving particle is m and the wavelength of the wave which is associated with its motion is λ, then the particle's speed is calculated from the relation; ..... .

(a) $v = \frac{h}{m\lambda}$ (b) $v = \frac{hm}{\lambda}$ (c) $v = \frac{\lambda}{hm}$ (d) $v = \frac{2h}{m\lambda}$

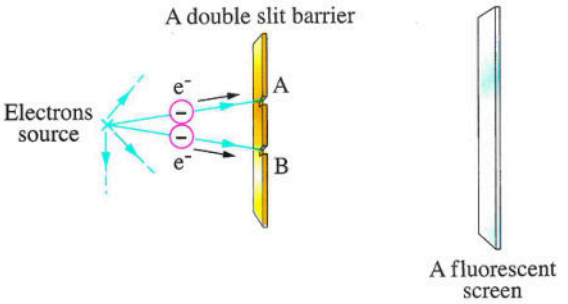
7 When an electron beam is directed towards a double slit as in the figure, ... will appear on the fluorescent screen.

(a) one bright spot at the middle

(b) two bright spots which are separated by a dark distance

(c) multiple bright and dark spots

(d) central dark spot surrounded with a bright circle



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48 The waves which are associated with the motions of a proton and an electron having the same wavelength, so ..... .

(Knowing that: mass of the proton > mass of the electron)

(a) the kinetic energy of the electron is less than that of the proton

(b) the momentum of the proton is greater than that of the electron

(c) the speed of the electron is greater than that of the proton

(d) the speed of the proton is greater than that of the electron

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49 Electron of mass m_e moves at a speed v, and the wavelength of the wave associated with its motion is λ. If the electron's speed is decreased to $\frac{v}{2}$, then the wavelength of the wave associated with its motion becomes ..... .

(a) 4λ (b) 2λ (c) $\frac{\lambda}{2}$ (d) $\frac{\lambda}{4}$

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50 If De-Broglie wavelength of the wave which is associated with the motion of an object of mass m is λ, then the kinetic energy of the object equals ..... .

(Where: h is Planck's constant)

(a) $\frac{2\,mh^2}{\lambda^2}$ (b) $\frac{\lambda^2}{2\,mh^2}$ (c) $\frac{h}{2\,m\lambda}$ (d) $\frac{h^2}{2\,m\lambda^2}$

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51 A light ray of wavelength 4500 Å has fallen on a metal surface, whose work function is 3.36×10^{-19} J, then:

(i) The maximum speed of the emitted electrons equals ..... .

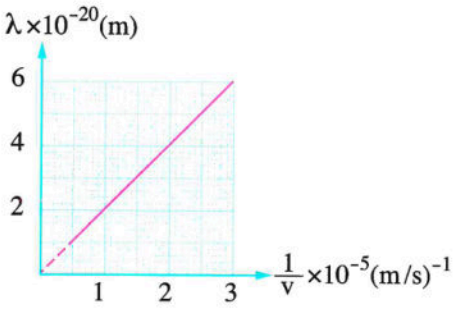
(a) 1.22×10^5 m/s (b) 2.34×10^5 m/s (c) 3.75×10^5 m/s (d) 4.82×10^5 m/s

(ii) The wavelength of the wave associated with the motion of the fastest emitted photoelectrons from the metal surface equals ..... .

(a) 1.5×10^{-9} m (b) 2.3×10^{-9} m (c) 5.8×10^{-9} m (d) 7.2×10^{-9} m

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52 The opposite graph represents the relation between De-Broglie wavelength of the wave associated with the motion of a body (λ) and the reciprocal of the speed of that body ($\frac{1}{v}$), then the mass of that body equals ..... .



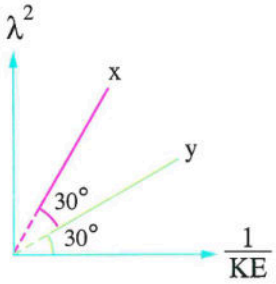
(Where: Planck's constant = 6.6×10^{-34} J.s)

(a) 1.3×10^{-19} kg (b) 2.3×10^{-19} kg

(c) 3.3×10^{-19} kg (d) 4.3×10^{-19} kg

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53 The opposite graph represents the relation between the square of wavelength (λ^2) of the wave associated with the motion of two particles x, y and the reciprocal of kinetic energy ($\frac{1}{KE}$), then the ratio between the masses of the two particles ($\frac{m_x}{m_y}$) equals ..... .



(a) $\frac{1}{3}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{3}{1}$ (d) $\frac{\sqrt{3}}{1}$

54 * The speed of an electron when a wave of wavelength 1Å is associated with its motion is ..... .

(a) 7.28×10^6 m/s (b) 2×10^7 m/s (c) 6.25×10^6 m/s (d) 1.37×10^7 m/s

55 * An insect moves with a speed of 12 m/s, if the wavelength of the wave associated with the motion of the insect is 5.5×10^{-30} m, then the mass of the insect is ..... .

(a) 10^{-4} kg (b) 1.2×10^{-4} kg (c) 2×10^{-5} kg (d) 10^{-5} kg

56 * A body of mass 10 kg moves with a speed of 5 m/s, then the wavelength of the wave which is associated with its motion is ..... .

(a) 1.325×10^{-30} m (b) 1.325×10^{-32} m (c) 1.325×10^{-35} m (d) 1.325×10^{-36} m

57 Two bodies x and y of masses m and 2 m are moving at speeds of v and 4 v, respectively, if the wavelength of the wave associated with the motion of body x is λ, the wavelength of the wave associated with the motion of the body y equals ..... .

(a) 8λ (b) 6λ (c) $\frac{\lambda}{6}$ (d) $\frac{\lambda}{8}$

58 Two objects have the same kinetic energy, if the wavelength of the wave that is associated with the motion of the first object is double the wavelength of the wave that is associated with the motion of the second object, the relation between the masses of the two objects m_1 and m_2 is .....

(a) $m_2 = \frac{m_1}{4}$ (b) $m_2 = \frac{m_1}{2}$ (c) $m_2 = 2\, m_1$ (d) $m_2 = 4\, m_1$

59 If a proton $\left({}_1^1H\right)$ and an alpha particle $\left({}_2^4He\right)$ are moving at the same speed, where the mass of the alpha particle equals 4 times the mass of the proton, the ratio between the wavelengths of the two waves associated with their motion $\left(\frac{\lambda_{proton}}{\lambda_{alpha}}\right)$ equals ..... .

(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{1}{4}$ (d) $\frac{4}{1}$

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60 The opposite table records the masses of three particles A, B and C which have the same type and amount of charge, by assuming that the three particles are accelerated by the same potential difference, then:

Particle	Mass (kg)
A	m
B	9 m
C	27 m

(i) The ratio of the kinetic energies which are gained by these particles is ..... respectively.

(a) 1 : 9 : 27 (b) 27 : 9 : 1 (c) 1 : 1 : 1 (d) 1 : 3 : 9

(ii) The particle whose motion has the largest wavelength is ..... .

(a) A (b) B (c) C (d) indeterminable

61 A static electron is accelerated under the effect of 2500 V, what is its maximum speed?

(a) 3×10^7 m/s (b) 2.5×10^8 m/s (c) 2.5×10^6 m/s (d) 1.5×10^8 m/s

62 * If the particle kinetic energy is increased to 16 times, the change in the percentage of De-Broglie wavelength is ..... .

(a) 25% (b) 50% (c) 60% (d) 75%

63 The opposite figure represents two bodies (x and y) of different masses and different kinetic energies, then the ratio of wavelengths for the waves that are associated with the motion of the two bodies $\left(\frac{\lambda_x}{\lambda_y}\right)$ is ..... .



(a) $\frac{4}{1}$ (b) $\frac{1}{4}$ (c) $\frac{1}{1}$ (d) $\frac{1}{3}$

64 In the cathode ray tube, when the potential difference between the anode and cathode is 700 V, the wavelength of the wave that is associated with the fastest electrons when they reach the anode equals ..... .

(a) 1.2×10^{-11} m (b) 2.3×10^{-11} m (c) 4.6×10^{-11} m (d) 6.4×10^{-11} m

65 In the electron microscope, the component responsible for focusing the electron beam is ..... .

(a) the anode (b) the cathode

(c) the filament (d) the magnetic lenses

66 In the electron microscope, which of the following represents the sequence of the results which occurs when the potential difference between the cathode and anode increases?

	The kinetic energy of electrons	The wavelength associated with the electron	The resolving power of microscope
(a)	increases	increases	increases
(b)	increases	decreases	decreases
(c)	increases	decreases	increases
(d)	decreases	decreases	decreases

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67 If the shortest wavelength that can be used in electron microscope is 1 Å, then:

(i) The maximum speed of the used electrons equals ..... .

(a) 7.28×10^6 m/s (b) 7.28×10^7 m/s (c) 3.28×10^5 m/s (d) 3.28×10^4 m/s

(ii) The potential difference between the anode and cathode equals ..... approximately.

(a) 662 V (b) 325 V (c) 442 V (d) 151 V

---

68 ✳ If the potential difference between the anode and cathode of an electron microscope is 500 V, the De-Broglie wavelength associated with the electron beam is ..... .

(a) 1.1×10^{-10} m (b) 5.49×10^{-11} m (c) 7.76×10^{-11} m (d) 4.14×10^{-12} m

---

69 ✳ In the electron microscope, an electron is affected by a potential difference of 20 kV, then:

(i) Its speed upon collision with the anode is ..... .

(a) 59.3×10^6 m/s (b) 83.9×10^6 m/s (c) 2.7×10^7 m/s (d) 7×10^7 m/s

(ii) The wavelength associated with its motion is ..... .

(a) 1.04×10^{-19} m (b) 1.04×10^{-16} m (c) 8.68×10^{-12} m (d) 2.7×10^{-10} m

---

70 An electron microscope is used to examine a particle twice. In the first time a potential difference of 16 kV is used while in the second time 25 kV is used, then the ratio between the maximum speeds of the electrons $\frac{(v_{max})_1}{(v_{max})_2}$ is equal to ..... .

(a) $\frac{3}{4}$ (b) $\frac{2}{3}$ (c) $\frac{4}{5}$ (d) $\frac{5}{7}$

---

71 An electron microscope is used to examine a particle twice. In the first time a potential difference (V_1) is used, so the momentum of the fastest electrons is (P_L) and in the second time a potential difference (V_2) is used, so the momentum of the fastest electrons is ($4\ P_L$), so the ratio between the two potential differences ($\frac{V_1}{V_2}$) equals ..... .

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{16}$ (d) $\frac{16}{1}$

---

72 In the electron microscope, three different potential differences V, 4V and 9V are used, then the ratio between:

(i) The maximum kinetic energy of electrons in the three cases, respectively, equals ..... .

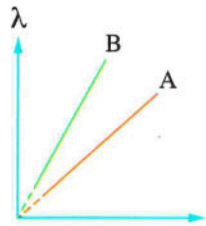
(a) 1 : 4 : 9 (b) 1 : 2 : 3 (c) $1 : \sqrt{2} : \sqrt{3}$ (d) 9 : 4 : 1

(ii) The wavelengths associated to the movement of electrons in the three cases, respectively, equals ..... .

(a) 1 : 4 : 9 (b) 6 : 3 : 2 (c) $1 : \sqrt{2} : \sqrt{3}$ (d) 1 : 2 : 3

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73 ✳ Two bodies A and B having the same charge are accelerated by the effect of a potential difference (V). The opposite graph represents the relation between the wavelength (λ) that is associated with the motion of the two bodies and the reciprocal of the square root of the accelerating voltage $\left(\frac{1}{\sqrt{V}}\right)$, then the relation between the masses of the two bodies is ..... .



(a) $m_A > m_B$ (b) $m_A < m_B$ (c) $m_A = m_B$ (d) indeterminable

Second Essay questions

1 **In Compton effect, what happens to each of the following physical quantities after collision compared to its value before collision:**

(a) Photon's speed.
(b) The wavelength of the photon.
(c) Linear momentum of the photon.
(d) Photon's energy.
(e) Equivalent mass of the photon.
(f) Electron's energy.
(g) Linear momentum of the electron.
(h) The wavelength associated to the motion of the electron.
(i) Mass of the electron.

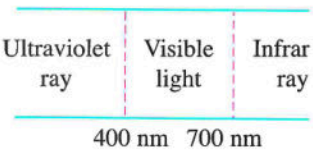
2 **Explain how** Compton phenomenon proves the particle nature of light (for photons).

3 A mass of 10 g, a part of it is converted in a nuclear reaction into energy of 9×10^{12} J, s **what is the remaining mass?**

4 A star emits energy at a rate 4.5×10^{25} J/s in space, **calculate** the mass that the star lose in one minute.

5 A single wavelength of electromagnetic radiation, the momentum of one of its photons is equal to the momentum of an electron with a speed of 2×10^5 m/s. **Calculate the wavelength of that radiation.**

6 **Using the opposite diagram, determine the electromagnetic spectrum region to which the photon belongs whose:**

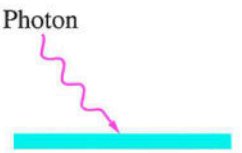


(a) Frequency is 4×10^{15} Hz.
(b) Equivalent mass is 4×10^{-36} kg.

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7 Two bodies (a and b), the kinetic energy of body (a) is four times the kinetic energy of body (b) and the ratio of the wavelength of the two waves associated to their motions is $\frac{2}{1}$ respectively, **find the ratio between** the masses of the two bodies $\left(\frac{m_a}{m_b}\right)$.

8 A photon of equivalent mass 3.2×10^{-36} kg has fallen on a metallic surface whose work function is 1.6 eV as in the opposite figure:



(a) **Show with explaining** if electrons are emitted from this surface.

(b) **What** is the equivalent mass of the photon falling on the surface so that an electron with maximum kinetic energy of 1.6 eV is emitted from it?

9 **What is the result of:** Falling of photons on a surface in which the interatomic spaces between its atoms are less than the wavelength of the photons?

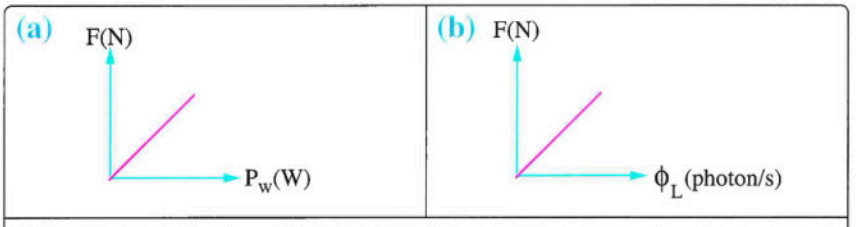
10 **What is the result of:** Falling of photons on a surface that the interatomic spaces between its atoms are greater than the wavelength of the photons?

11 A monochromatic electromagnetic beam of power 200 W and the wavelength of its photons 8×10^{-7} m has fallen perpendicularly on a reflecting surface, **calculate:**

(a) The linear momentum of the photon of that beam.

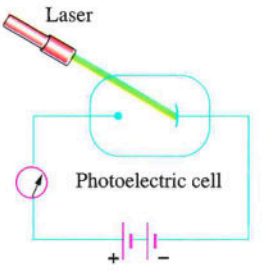
(b) The force by which the beam acts on this surface.

12 **Write the mathematical relation and the equivalent to the slope of the straight line for each of the following graphs:**



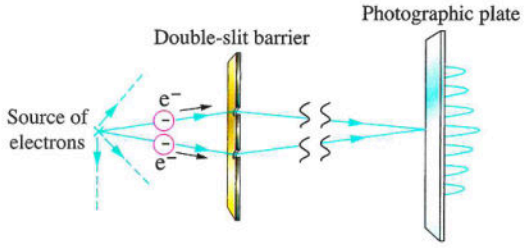
Where: (F) is the force by which a light ray acts on a reflecting surface, (P_w) is the power of the light ray, (ϕ_L) the rate of the falling photons.

13 The opposite figure shows a photoelectric cell such that a laser ray of wavelength 550 nm and power 0.2 mW is incident on its cathode. If the longest wavelength required for the emission of electrons from this cathode is longer than 550 nm and assuming that every incident photon makes an electron get emitted and all emitted electrons reach the anode, **calculate** the maximum photocurrent intensity in the circuit.



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14 The opposite figure illustrates an electron beam being incident on a photographic plate after passing through a double-slit, **explain why** the pattern formed by the electron beam on the photographic plate supports the wave nature theory of particles.



15 A single wavelength of electromagnetic radiation, its photon's energy is 4 eV, falls on a cathode of photoelectric cell whose work function is 2 eV. **Calculate** the shortest wavelength of the wave associated with emitted electrons.

16 A free electron is accelerated from rest by the effect of a potential difference of 1000 V, **calculate** De-Broglie wavelength associated with the motion of that electron when reaching the anode.

17 **Compare between:** The electron and the photon (in terms of: nature - rest mass - momentum - acceleration possibility).

18 **Mention the condition for:** Observing the details of the structure of a tiny body by using the microscope.

19 **Explain why** the optical microscope can't be used to see the details of Corona virus.

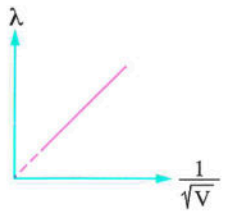
20 When increasing the potential difference between the anode and cathode in an electron microscope, **what happens with explanation for each of:**

(a) The maximum kinetic energy of the electrons that reach the anode.

(b) The wavelength of the wave associated with the motion of electrons.

(c) The resolving power of the microscope.

21 The opposite graph represents the relation between the wavelength (λ) associated with the electron radiation in electron microscope and the reciprocal of the square root of the potential difference between the cathode and anode $\left(\frac{1}{\sqrt{V}}\right)$. **What** is the magnitude of the slope of the graph line?



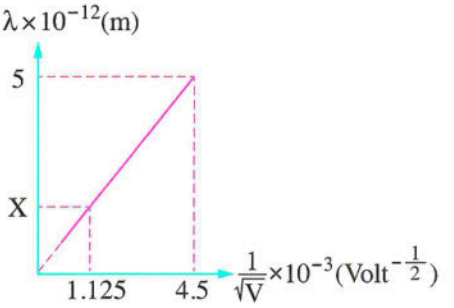
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Questions of the Previous Years Exams on Lesson 2

1 In Compton effect, when a photon of γ-ray collides with a free electron moving with velocity v, so ..... . *(Exp. June 21)*

	The momentum of the electron after collision	The momentum of the scattered photon
(a)	increases	increases
(b)	decreases	decreases
(c)	increases	decreases
(d)	decreases	increases

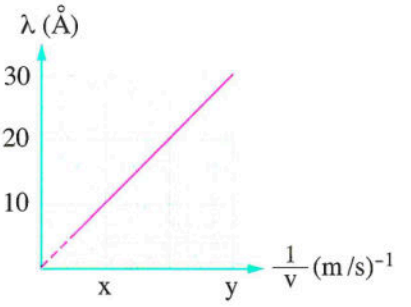
2 The graph represents the relation between the reciprocal of the square root of the electric potential difference used in the cathode ray tube and the wavelength associated with the electrons emitted from the tube filament when reaching the anode, so the value of point (X) on the graph equals ..... .



(a) 1.25×10^{-12} m (b) 2.5×10^{-12} m

(c) 2×10^{-11} m (d) 1.5×10^{-11} m *(Exp. June 21)*

3 The graph represents the relation between the wavelength (λ) for the electrons emitted from the tube filament and the reciprocal of the velocity $\left(\frac{1}{v}\right)$ of the electrons, so the ratio of $\frac{\text{The velocity of the electron at point x}}{\text{The velocity of the electron at point y}}$ = ...



(Take: $h = 6.625 \times 10^{-34}$ J.s, $m_e = 9.1 \times 10^{-31}$ kg) *(Exp. June21)*

(a) $\frac{9}{1}$ (b) $\frac{1}{9}$ (c) $\frac{3}{1}$ (d) $\frac{1}{3}$

4 An electron microscope is used to examine two different viruses (x) and (y).

If the dimension of virus (x) is 1 nm while that of virus (y) is 4 nm, the ratio of

$$\frac{\text{The required potential difference between anode and cathode to observe virus (x)}}{\text{The required potential difference between anode and cathode to observe virus (y)}} = \text{.....}$$

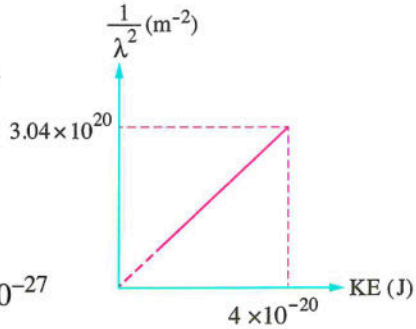
(a) 16 (b) 8 (c) 4 (d) 2 *(Exp. June 21)*

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5 In Compton effect, when a photon of γ-ray collides with a free electron moving with velocity v, so ..... . *(1st session 21)*

	The mass of the electron	The wavelength of the photon
(a)	doesn't change	decreases
(b)	decreases	decreases
(c)	doesn't change	increases
(d)	increases	decreases

6 The opposite graph represents the relation between the reciprocal of the square of the wavelength $\left(\frac{1}{\lambda^2}\right)$ associated with a particle motion and its kinetic energy (KE), so the mass of the moving body equals ..... kg.



(1st session 21)

(a) 1.67×10^{-27} (b) 3.33×10^{-27}

(c) 7.6×10^{-39} (d) 3.8×10^{-39}

7 A moving body has a mass of 140 kg, the wavelength of the wave associated with its motion equals 1.8×10^{-34} m. Given that, Planck's constant equals 6.625×10^{-34} J.s, so the velocity of the body equals ..... m/s. *(1st session 21)*

(a) 2.629×10^{-3} (b) 2.269×10^{-3} (c) 0.26×10^{-3} (d) 26.29×10^{-3}

8 In the electron microscope, when the potential difference between the cathode and the anode is increased from 25 kV to 100 kV, the wavelength associated with the motion of the electron beam ..... . *(1st session 21)*

(a) decreases to its half (b) increases to its double

(c) decreases to its quarter (d) increases to its quadruple

9 In Compton effect, when a photon of X-ray collides with an electron moving with velocity (v), ... *(2nd session 21)*

	The speed of the electron after collision	The equivalent mass of the photon after collision
(a)	increases	increases
(b)	increases	decreases
(c)	decreases	decreases
(d)	decreases	increases

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10 Assuming that the velocity of electron of mass 9.1×10^{-31} kg equals to the velocity of proton of mass 1.67×10^{-27} kg, so the wavelength of the wave associated with the electron motion equals ..... of that associated with the proton motion. *(2nd session 21)*

(a) 545 times (b) 1545 times (c) 1835 times (d) 835 times

11 If the energy of the photon that is used in the optical microscope = 496.88×10^{-21} J and the momentum of the moving electron in the electron microscope = 7.626×10^{-23} kg.m.s^{-1}, a body of dimension 400 nm can be seen using ..... . *(2nd session 21)*

(a) the optical microscope only (b) both optical and electron microscopes
(c) the electron microscope only (d) the naked eye only

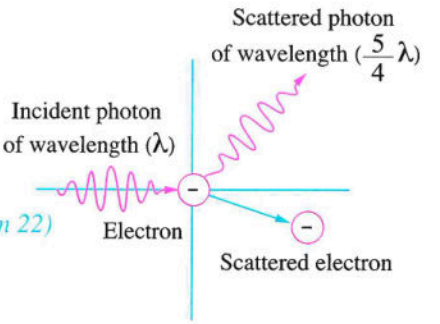
12 An electron microscope is used to examine two different viruses (A) and (B), and the collected data are recorded in the following table:

The virus	Its dimension (diameter)	The applied potential difference between anode and cathode to examine the virus
A	10 nm	1.5 kV
B	X	37.5 kV

By using these data, the value of (X) equals ..... . *(2nd session 21)*

(a) 1 nm (b) 0.4 nm (c) 0.8 nm (d) 2 nm

13 The figure illustrates an X-ray photon colliding with a free electron, where the data that describe the photon before and after scattering are shown in figure, so the incident photon has lost ..... of its original energy as a result of that collision. *(1st session 22)*



(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{1}{5}$ (d) $\frac{4}{5}$

14 A moving photon has equivalent mass (3.68×10^{-38} kg), then its wavelength is equal to ..... . *(1st session 22)*

(a) 40 μm (b) 50 μm (c) 30 μm (d) 60 μm

15 Photon (x) has wavelength 320 nm and photon (y) has wavelength 240 nm, so the ratio between the momentum of photon (x) to that of photon (y) $\left(\frac{P_{L(x)}}{P_{L(y)}}\right)$ = ... *(1st session 22)*

(a) $\frac{4}{3}$ (b) $\frac{3}{4}$ (c) $\frac{4}{1}$ (d) $\frac{3}{1}$

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16 An electron microscope is used to magnify a virus of a dimension X with using a potential difference of V. If the virus is replaced with another of a dimension $\frac{1}{10}$ X, the potential difference should be increased by ..... . *(1st session 22)*

(a) 100 V (b) 9 V (c) 99 V (d) 10 V

17 In Compton phenomenon, it was noticed that when a γ-ray photon of wavelength λ falls on a free electron, the photon lost $\left(\frac{1}{4}\right)$ of its original energy, so the wavelength of the scattered photon becomes ..... . *(2nd session 22)*

(a) 4λ (b) $\frac{4}{3}\lambda$ (c) $\frac{3}{2}\lambda$ (d) 2λ

18 Photon (x) has frequency 9.375×10^{14} Hz and photon (y) has frequency 1.25×10^{15} Hz. So, the ratio between the momentum of photon (x) to that of photon (y) $\frac{P_{L(x)}}{P_{L(y)}}$ equals ..... *(2nd session 22)*

(a) $\frac{4}{3}$ (b) $\frac{4}{1}$ (c) $\frac{3}{1}$ (d) $\frac{3}{4}$

19 A photon has frequency 7.9×10^{11} kHz, then its equivalent mass at motion equals ..... *(2nd session 22)*

(a) 5.82×10^{-39} kg (b) 1.74×10^{-27} kg (c) 5.82×10^{-36} kg (d) 1.74×10^{-30} kg

20 In electron microscope, the ratio between maximum speed of electrons when using potential difference of 60 kV to its maximum speed when using potential difference of 20 kV equals ..... . *(2nd session 22)*

(a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) 3 (d) $\frac{1}{3}$

21 In Compton effect, when a photon of gamma ray collides with a free electron. Which of the following is correct regarding the photon as a result of the collision? *(Exp. 23)*

	Linear momentum of the scattered photon	Wavelength of the scattered photon
(a)	Decreases	Constant
(b)	Increases	Decreases
(c)	Decreases	Increases
(d)	Increases	Increases

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22 Two photons (X) and (Y) propagate in air, the frequency of (X) is more than that of (Y). Which of the following choices is correct? *(Exp. 23)*

(a) The velocity of photon X is less than that of Y.
(b) The energy of photon X is less than that of Y.
(c) The wavelength of photon X is more than that of Y.
(d) The momentum of photon X is more than that of Y.

23 If the wavelength of red light is the greatest wavelength in the visible light, which of the following choices is correct? *(Exp. 23)*

(a) The frequency of the photons in red light is the greatest in visible light.
(b) The energy of the photons in red light is the greatest in visible light.
(c) The momentum of the photons in red light is the smallest in visible light.
(d) The speed of photons in red light in air is greater than that of visible light.

24 The resolving power of the electron microscope is high, because the accelerated electrons have ..... . *(Exp. 23)*

(a) high kinetic energy and short wavelength associated to their motion
(b) high kinetic energy and long wavelength associated to their motion
(c) low kinetic energy and short wavelength associated to their motion
(d) low kinetic energy and long wavelength associated to their motion

25 A photon has frequency (4.2×10^{14} Hz), then its linear momentum is equal to ..... . *(1st session 23)*

(a) 9.275×10^{-26} kg.m/s
(b) 9.275×10^{-28} kg.m/s
(c) 9.275×10^{-30} kg.m/s
(d) 9.275×10^{-24} kg.m/s

26 A cathode ray tube works at a potential difference 2000 V, while another identical tube works at a potential difference 8000 V, then the ratio between the wavelength of the associated wave to the emitted electrons from the first tube cathode and the wavelength of the associated wave to the emitted electrons from the second tube cathode $\left(\frac{\lambda_1}{\lambda_2}\right)$ equals ..... .

(a) $\frac{2}{1}$ (b) $\frac{4}{1}$ (c) $\frac{6}{1}$ (d) $\frac{8}{1}$ *(1st session 23)*

27 In order to examine a virus of 20 nm dimensions in the electron microscope, a potential difference (V) is used, so in order to examine another virus of dimensions 15 nm, the potential difference must be ..... . *(1st session 23)*

(a) increased by 0.78 V
(b) decreased by 0.78 V
(c) increased by 0.5 V
(d) decreased by 0.5 V

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28 A proton moves at a velocity of 3×10^6 m/s, so the wavelength of its accompanied wave is ..... . ($m_p = 1.67 \times 10^{-27}$kg) *(2nd session 23)*

(a) 1.32×10^{-13} m (b) 7.5×10^{-14} m (c) 1.32×10^{-10} m (d) 7.5×10^{-10} m

29 X-ray photon of wavelength (4×10^{-9} m) lost 0.1% from its energy when it collided with a static electron, so the wavelength of the scattered photon after collision becomes ..... . *(2nd session 23)*

(a) 3.996×10^{-9} m (b) 4.002×10^{-9} m (c) 4.004×10^{-9} m (d) 4.008×10^{-9} m

30 A potential difference is applied in an electron microscope to give the electrons a velocity of 1.8×10^7 m/s in order to see a virus of length 300 Å. Knowing that Planck's constant $= 6.625 \times 10^{-34}$ J.s and the mass of electron $= 9.1 \times 10^{-31}$ kg, so ..... . *(2nd session 23)*

	The wavelength of the electrons waves	The vision
(a)	0.4 Å	can't be seen
(b)	0.4 Å	can be seen
(c)	4 Å	can be seen
(d)	4 Å	can't be seen

31 A photon has energy hυ, so its momentum and wavelength equal ..... .
(Knowing that h is Planck's constant, v is photon's frequency) *(1st session 24)*

	Momentum	Wavelength
(a)	$\frac{h\upsilon}{c}$	$\frac{\upsilon}{c}$
(b)	$\frac{h\upsilon}{c}$	$\frac{c}{\upsilon}$
(c)	$\frac{c}{h\upsilon}$	$\frac{\upsilon}{c}$
(d)	$\frac{c}{h\upsilon}$	$\frac{c}{\upsilon}$

32 An object of dimension X is needed to be seen by an optical microscope. So the minimum momentum of the photon used in the light beam can be equal to ..... . *(1st session 24)*

(a) $\frac{h}{3X}$ (b) $\frac{h}{X}$ (c) $\frac{3h}{X}$ (d) $\frac{X}{h}$

33 Photons fall vertically on a metallic surface with a rate of (ϕ_L), if the energy of the photon = hυ, so the change in the momentum of one photon due to its reflection = ... *(2nd session 24)*

(a) $\frac{2h}{\upsilon}$ (b) $\frac{h\upsilon}{2c}$ (c) $\frac{h\upsilon}{c}$ (d) $\frac{2h\upsilon}{c}$

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34 A photon of energy 1.77×10^3 eV, so its momentum = ... *(2nd session 24)*

(a) 9.44×10^{-25} kg.m/s (b) 9.44×10^{-15} kg.m/s

(c) 5.9×10^{-6} kg.m/s (d) 8.496×10^{-8} kg.m/s

35 A photon of frequency υ falls on a metallic surface that has critical frequency of $\frac{\upsilon}{2}$ so that an electron is liberated from the surface of the metal by speed v and another photon of frequency 2 υ falls on the same metallic surface, then the speed of the liberated electron in the second case equals ..... . *(2nd session 24)*

(a) $\sqrt{5}$ v (b) $\sqrt{3}$ v (c) 2 v (d) $\sqrt{6}$ v

36 If a potential difference of 300 V is applied between the anode and cathode in an electronic microscope. Which choice is correct? *(2nd session 24)*

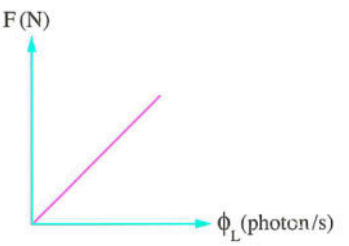
	The wavelength of the wave accompanied to the electron motion	The max. speed of the released electron
(a)	7.09 Å	1.027×10^7 m/s
(b)	0.07 nm	1.027×10^7 m/s
(c)	0.07 Å	1×10^8 m/s
(d)	7.09 nm	1×10^8 m/s

37 After the collision of a photon by a free electron in Compton's effect, ... decreases. *(Practice 25)*

(a) electron's speed (b) electron's energy

(c) photon's speed (d) photon's frequency

38 A beam of light of single wavelength falls perpendicular to a reflecting surface and the opposite graph represents the relation between the force (F) that acts on the surface when it is reflected and the rate of falling photons of the beam on the surface (ϕ_L), so the slope of the graphical line represents ..... . *(Practice 25)*



(a) photon's energy (b) photon's frequency

(c) double the momentum of photon (d) half the momentum of photon

39 A photon whose energy is 4 eV, so the equivalent mass of photon equals ..... . *(Practice 25)*

(a) 9.22×10^{-36} kg (b) 8.33×10^{-36} kg (c) 7.11×10^{-36} kg (d) 5.44×10^{-36} kg

40 The incident light beam on a lustrous surface causes ..... on the surface. *(Practice 25)*

(a) force only (b) pressure only

(c) force and pressure (d) no force or pressure

41 If the kinetic energy of a body increases 16 times, the percentage of change in the wavelength of De-Broglie is ..... . *(Practice 25)*

(a) 25% (b) 50% (c) 75% (d) 100%

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42 The idea of working of the electron microscope depends on ..... . *(Practice 25)*

(a) the wave nature of electrons (b) the particle nature of electrons

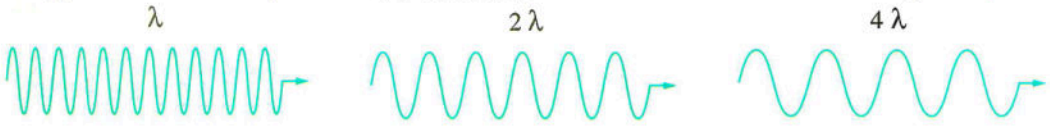
(c) the wave nature of photons (d) the particle nature of photons

43 When a beam of photons strikes a reflective surface vertically at a rate Y photons/s, and the momentum of each photon is X, then the force excerted on the surface equals ..... .

(Practice 25)

(a) XY (b) 4 XY (c) 2 XY (d) 0.5 XY

44 The figure shows three photons (1) , (2) , (3) each labeled with its wavelength. *(Practice 25)*



(1) **(2)** **(3)**

If the energy of photon (3) is 3 eV, then the energy of photon (1) is greater than that of photon (2) by .....

(a) 6 eV (b) 4 eV (c) 8 eV (d) 9 eV

45 In Compton effect, what causes the wavelength of X-ray photon to increase when it collides with a free electron? *(1st session 25)*

(a) The photon gains some energy from the electron.

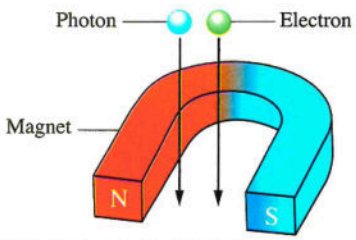
(b) The photon loses some of its energy to the electron.

(c) The photon speed decreases as a result of its collision with the electron.

(d) The photon momentum increases as a result of its collision with the electron.

46 Assume that an electron and a photon move in a straight line to the region between the poles of a magnet as shown in the figure.

What would happen to the path of each of them? *(1st session 25)*



	Electron	Photon
(a)	Deviates from its straight path	Deviates from its straight path
(b)	Continues on its straight path	Deviates from its straight path
(c)	Deviates from its straight path	Continues on its straight path
(d)	Continues on its straight path	Continues on its straight path

47 If electrons are accelerated in an electron microscope till their energy reached 10^4 eV, what will be the shortest wavelength of the wave associated with the used electrons? *(1st session 25)*

(a) 12 Å (b) 0.012 Å (c) 1.2 Å (d) 0.12 Å

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48 A photon of gamma rays having wavelength of (λ_1) collided with a free electron whose motion was accompanied by a wavelength of (λ_2), and both were scattered.

The results of this collision include ..... . *(2nd session 25)*

	Wavelength of photon (λ_1)	Wavelength associated with the electron (λ_2)
(a)	increases	increases
(b)	decreases	decreases
(c)	increases	decreases
(d)	decreases	increases

49 Two photons, A and B, of spectra having wavelengths of 800 nm and 200 nm, respectively.

The ratio of momenta of the two photons $\left(\frac{(P_L)_A}{(P_L)_B}\right)$ is equal to ..... . *(2nd session 25)*

(a) $\frac{1}{4}$ (b) $\frac{4}{1}$ (c) $\frac{1}{2}$ (d) $\frac{1}{1}$

50 In the electron microscope, as the potential difference applied to accelerate the electron beam is increased from (X) to (4 X).

So, the wavelength associating the electron beam will ..... . *(2nd session 25)*

(a) decrease to quarter (b) increase to four times

(c) decreases to half (d) increases to double

51 A monochromatic light beam of wavelength 4500 Å falls on the surface of a metal as a result photoelectrons are emitted, if the power of this incident light beam is 10 W.

Calculate the maximum rate of the photoelectrons emission from the surface of the metal. *(2nd session 23)*

.....

52 A radio station whose power is 100 kW transmits on a wave with a frequency of 92.4 MHz, if Planck's constant equals 6.625×10^{-34} J·s, **calculate:** *(Practice 25)*

(a) the energy of an emitted photon.

(b) the number of photons emitted in one second.

.....

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